

A Perspective on Bidirectional Human-AI Alignment and Human Autonomy

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Abstract

We argue that the values in the Bidirectional Human-AI Alignment framework need to be further unpacked, and that interaction design needs to be considered in relation to supporting such values. AI alignment with human values is often operationalized as promoting understanding of AI systems and aligning model behavior. However, an interactional perspective foregrounds additional ways of supporting human autonomy. We illustrate this by describing understandings of the value of human autonomy and (interaction design) methods to support it. We provide an overview of different strategies for supporting human autonomy in design and use of AI systems, with consideration of material and experiential aspects, different stakeholders, and different lifecycle phases.

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models; **Interaction design theory, concepts and paradigms.**

Keywords

Autonomy, Agency, AI Alignment, Bidirectional Alignment, Interaction Design, Liberty

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1 Bidirectional Human-AI Alignment

The AI alignment problem asks the question which human values AI systems should align with, who gets to decide, and how this can be achieved. Gabriel [13] argues that AI could be aligned with values such as human rights or AI ethics principles. Shen et al. [24] adopt human values as the alignment objective, and emphasize the bidirectional nature of the human-AI alignment problem. On the one hand, AI systems should be aligned with humans by training AI models in accordance with human specifications. On the other hand, they propose to “Align Humans to AI” by supporting humans in understanding, critiquing, and collaborating with AI. Their work

provides a good starting point; yet it needs to be extended to unpack its full potential, in particular regarding the underlying values.

We claim that the discourse of AI alignment would benefit from more theoretical clarity regarding the involved key concepts. While the goal of (bidirectional) alignment is strong and clear from a high-level perspective, involved value concepts like autonomy, fairness, or benevolence (Sec 3.1 in [24]) carry a lot of normative weight while remaining conceptually broad. We argue that systematically addressing them in AI system design requires better operationalization from both a theoretical and a methodological perspective. This would allow for better support across stakeholders and system lifecycle phases. Dimensions in the typology by Shen et al. [24] place a strong emphasis on AI model feedback and AI literacy. However, supporting human autonomy specifically requires enabling meaningful, value-aligned action at different lifecycle phases. The framing “Align Humans to AI” implies that humans need to recalibrate, learn, and adjust to AI, while in practice, humans already adapt to technical systems. From an HCI perspective, the description of this side in the Bidirectional Human-AI Alignment framework [24] is strong on evaluation and cognitive support in understanding AI systems, but weaker when it comes to identifying what AI value alignment means for interaction design and power asymmetries. Fortunately, the HCI community has investigated many methods to this end, including methods such as participatory design.

In this short position paper, we illustrate the application of the Bidirectional Human-AI Alignment framework on one of the key values: human autonomy. Human autonomy has been described as one of the fundamental HCI concepts, even a potential boundary object for HCI research [5]. We (1) discuss our recent theoretical contributions by conceptualizing human autonomy through the lens of positive and negative liberty, and provide a scale for its measurement; (2) expand bidirectional alignment by distinguishing material and experiential dimensions, and (3) map design strategies related to human autonomy across AI system lifecycle phases and stakeholders.

2 Human Autonomy: Theory and Measurement

Human autonomy is complex to unpack, as it can be understood as both an intrinsic value in itself and in terms of value-aligned action, opening the door for other values. Moreover, it is described as closely related with agency [5].

Strategies	Material (Objective) impacts on human autonomy	Experiential (subjective) impacts on human autonomy
End user and others directly affected/involved in system use. <i>Understanding how AI system makes decisions, affects user, and how value-aligned action is possible [15], consent to use, inspect what the system is doing, and perform meaningful action.</i>		
<i>Understanding vs. acting</i>	Transparency makes clear how data will be used, thus enabling meaningful choices, fostering human autonomy [10] (e.g., <i>communicate what a user can do with a system, how well it functions, on what basis it makes decisions [2]</i>). Related: explainable AI (XAI). When provided with explanations on AI system limitations, this can augment agency through expanding a user's possibilities of action (e.g., people are informed on actions they can take, can contest decisions, or appropriate the system for their own needs [9]). For example, end users and others affected by use of the system may authorize data use through informed consent [10].	Transparency also has an experiential component. Transparency may make it more clear how data is used, while at the same time not providing any meaningful way to act on the system. Moreover, some people are more capable and willing to understand technical explanations than others, and some XAI techniques have been found to create overtrust rather than appropriate reliance [9]. Information on AI systems may be hidden to improve UX. Clearer principles regarding when and how to address transparency and explainability in practice are needed [14].
<i>Constraints</i>	The system does not impose constraints that impede on value-aligned action (cf. negative liberty [25]). Imposing constraints can in some cases promote human autonomy in the long term (e.g., digital self-control) [5].	A person interaction with the system experiences no constraints (cf. perception of negative liberty [25]).
<i>Value-aligned action</i>	The system enables value-aligned action (cf. positive liberty [25]), meaning they have options that align with their personal goals. The primary user has the possibility to preserve choices, or the system directly enables the possibility for value-aligned action by others affected by system use.	The user has the perception that the way the system works aligns with their personal goals (cf. perception of positive liberty [25]).
<i>User control</i>	User control : User actions impact the system (e.g., <i>supporting invocation, dismissal and correction, giving user controls to customize AI system behavior [2], end-user programming</i>).	Giving users as much control as possible may seem like a way to support human autonomy, but <i>excessive</i> information and choice may actually hinder achievement of a user's higher-level goals [11]: " <i>autonomy is protected when users are given control over the right things at the right time</i> " [11, p.18], which is shaped through design. Another experiential aspect of user control can be realized through system feedback : a person interacting with the system feels their actions directly impact the system.
<i>Adaptation</i>	The system learns from human behavior and explicit feedback, and adapts to human(s) [2].	A person interacting with the system is aware of adaptation .
<i>Participation</i>	Choice to opt-out of system use by primary user and those affected by use.	Giving user space for reflection (on meaningful action).
Society <i>Supporting society at large in acting with AI systems</i>		
<i>Understanding vs. acting</i>	Human autonomy has been interpreted as maintaining human self-determination and ability to take part in democratic processes [15]. Supporting democratic choice regarding regulation of AI systems, and in what ways (and in which directions) AI development should be fostered through public funding and regulation.	Promoting AI literacy throughout the population, so that people understand how value-aligned action is possible.
Polymakers, regulators, controllers		
<i>Regulation & Monitoring</i>	Enabling legal action. Ethics-based AI auditing through continuous evaluation of system outputs and performance against ethics principles, conducted by government officials, within an organization, or by a third party [19].	For public sector AI applications, legal systems and private companies exert pressures that shape design, adoption, and use decisions. Experienced power imbalances can lead to frustration when these limit the ability of public sector actors to address ethical issues [17]. Creation of strong incentives may help address such power imbalances (e.g., through liability of AI system vendors, legal requirements for certification, and support for more power-balanced procurement contracts) [17].

Table 1: Strategies for promoting human autonomy (and agency) for different (human) stakeholders involved in AI systems use. We distinguish material (objective) impacts from experiential (subjective) impacts. (Table is continued on the next page.)

Strategies	Material (Objective) impacts on human autonomy	Experiential (subjective) impacts on human autonomy
AI System Designer/Developer <i>Enabling inspection of the functioning of the AI system and outcomes.</i>		
<i>Explainability, Observability</i>	Tracing system functioning and changing it. XAI methods can be used to generate explanations of system behavior, which also supports communication with stakeholders. Observability : detection of data, model, and concept drift, and prevention of incidents, using a variety of comprehensive, system-generated metrics, logs, traces, and events [4].	
<i>Awareness and prioritization of ethics issues in AI development</i>	Human-centered AI development may not be reflected in software development practice in some companies due to a lack of understanding of ethics issues, or assigning higher priority to other issues. A capability maturity model could be used to evaluate this for human-centered AI development [14].	Developers may encounter <i>moral stress</i> when they identify ethical issues in their work but lack the capacity to address these [22]. Such situations can cause stress, and the mismatch can result in failure to act in accordance with their values [22].

Table 2: Table 1 continued.

Phase	Strategy and (material) impact
AI system <i>Align system behavior for human autonomy</i>	
During design	
<i>Participation</i>	Including all affected stakeholders as active actors during the system design/development stages instead of addressing issues post factum [1, 20, 27]. Participatory design is a method that aims to include needs, preferences and values of stakeholders in the design process, empowering stakeholders in AI systems design. In practice, participatory approaches in AI design projects are often limited to providing preferences in a consultative way, instead of having an impact on the scope and purpose of a system, or the choice whether AI should be used at all, limiting agency of stakeholders [8]. Power and knowledge differentials exist between members of the community and, e.g., decision-makers in public sector agencies. Training on working effectively with community members, awareness of participatory methods, and incentives to use them are beneficial. Participatory AI methods themselves may need to be further developed to be better suited to addressing problems in practice, which requires social and political work, and resources [17].
<i>Integrating values</i>	Value-Sensitive Design is a technology design approach that integrates accounting for human values into the design process. It takes an interactional perspective, where people, social systems, and technologies mutually affect each other. It uses conceptual, empirical, and technical lenses to analyze values, how these are supported through specific technologies, tradeoffs between values, and success of designs [12].
<i>Anticipation of harm</i>	Chan et al. [7] provided four potential harm characteristics that make it harder to trace responsibility back to any single actor: underspecification, directness of impact, goal-directedness, and long-term planning. Anticipating the harms and working for the opposite effect can help to minimize the problem. Seamful design for XAI involves identifying potential harms and mismatches between design assumptions and deployment reality across the system’s lifecycle, and making AI imperfections visible to end users in order to support user agency [9].
During data collection and model training	
<i>Human feedback</i>	Giving feedback to, e.g., samples of behavior in reflection to preferences held by the individual (e.g., fine-tuning LLMs with pairwise preferences). For RLHF and interactive RL, taxonomies and lists of feedback types have been developed [3, 16, 18]. The focus in such taxonomies tends to be on the information quality of human feedback (e.g., is feedback consistent and usable for training). Where human-centered aspects of feedback are discussed, the focus tends to be on promoting high information quality (e.g., ease of use, cognitive load). Values remain implicit as part of human preferences. Effects of single feedback actions are minimal, while the combination of many instances across the entire population will display biases. To investigate how this enables human autonomy, we need to consider to what extent giving feedback to an AI system can be understood as meaningful action, and to what extent aggregated feedback leads to a value-aligned result.
During deployment	
<i>Human oversight</i>	Human-in-the-loop , human oversight [15]. Overriding system decisions to prevent adverse outcomes for self/others. Observability can be enhanced through continuous monitoring (of metrics, logs, uncertainties), integration of explainability tools (e.g., LIME, SHAP) in the monitoring dashboard, and independent overseeing agents [4]. These measures need to be integrated during the system design and development stages.

Table 3: Strategies for human autonomy with a focus on material impacts on AI systems, for different lifecycle phases.

In the Bidirectional Human-AI Alignment framework, which builds on Schwartz’s Theory of Basic Values (and which lists autonomy as an interactional requirement), autonomy is a value that is part of Self Direction [23, 24]. Human autonomy is not only a central concept in HCI, but it is also an important principle for AI governance. The EU Ethics Guidelines for Trustworthy AI list human autonomy as a core requirement [15]. Such policies articulate strong normative commitments but remain largely high-level, with their most concrete guidance centering on data governance and human understanding and control of system behavior.

In the HCI literature, human autonomy has been treated as having intrinsic value in itself, as a basic human need (for example, in Self-Determination Theory), or alternatively as providing a benefit to users or other stakeholders [5, 21]. Views on autonomy (and agency) often include an element of causality (whether the user is causally involved in execution and/or decision-making) or elements of identity (user’s self and values) [5]. Most recently [25], we have described how the concepts of autonomy and agency relate, and proposed the concepts of positive and negative liberty as structural conditions for them. Adapted from ideas in political philosophy [6], we described *negative liberty* as freedom from constraints to action, while *positive liberty* represents freedom to act in a value-aligned way. Together, positive and negative liberty are prerequisites for agency. **Agency** refers to the capacity to act, with an experience of felt authorship of actions. **Autonomy** can emerge when the structural conditions of positive and negative liberty are fulfilled, and a person can exercise agency in a value-aligned way.

This distinction is not only of theoretical value, but it also allows us to measure its sub-components, supporting the evaluation of design aspects. Positive liberty was assessed through scale components such as “[The way the AI system works] is consistent with my values and priorities” (where “the AI system” was replaced by the specific functionality), whereas negative liberty was assessed through items such as “I have complete freedom to manage this situation without external obstacles”. This allows for assessing the *perception* of liberty [25]. In systems design, objective liberty would refer to actual freedom to act in a value-aligned way, and the factual absence of obstacles. There are thus both subjective and objective components to liberty and autonomy. This is closely connected to the distinction between experiential (subjective) and material (objective) aspects of human autonomy. Bennett et al. [5] raise the question how material and experiential aspects of human autonomy in interactive systems design can be disentangled. Material influence refers to having an actual, objective, real-world impact over outcomes when interacting with the system, while experiential aspects refer to the feeling of control (for instance, sense of agency [26]). That is, hypothetically, autonomy may be increased experientially without materially increasing it [5] if an interface gives an end user the feeling of choice, without them having an actual impact on system decision-making. For promoting human autonomy in system design, we argue it is crucial to consider both factors. As both perception of the possibility to act and actually allowed actions by the system impact human behavior, we argue that the loop needs to be closed for operationalization.

3 Strategies to Foster Human Autonomy

We provide a (non-exhaustive) overview of existing strategies to foster human autonomy and agency, from a perspective that considers different human stakeholders and distinguishes material from experiential aspects (Tables 1-2) and from an AI systems perspective that considers different phases (Table 3). Multiple stakeholders are affected by AI systems (cf. [10]).

As interaction design needs to be considered in the context of Bidirectional Human-AI Alignment, we discuss specific interaction design strategies and methods. In Table 1, we elaborate on strategies to foster human autonomy at the level of the end user, at the society level, for AI system designers, policymakers, and controllers. At the user level, we incorporate existing user-centric design guidelines for Human-AI interaction [2]. What is important to keep in mind regarding interaction design strategies to foster user autonomy, is that multiple stakeholders can be affected by the same (decision) process, who may or may not be direct “users” of the system. These different stakeholders can hold different sets of values. Supporting agency and autonomy for multiple stakeholders means preserving choice and information for these different stakeholders. For example, a diagnosis interface by a doctor may impact patient autonomy. It should be clear that these stakeholders interact (see e.g., [9] who argue human-centered XAI uses a sociotechnical perspective) and that their goals and preferences are not static (cf. [11]). Some of the methods have a focus on fostering human understanding of how AI systems work, for example through transparency and interpretability (AI system designer level). Other methods specifically focus on interaction design (e.g., user control and feedback). While some listed approaches are not HCI methods per se, we include perspectives on how autonomy may be supported at multiple levels, including society and policymakers, to emphasize a sociotechnical approach to AI alignment. At the AI system level (Table 3), we look at methods that have been discussed in HCI for supporting human autonomy during different lifecycle phases.

4 Conclusion and Research Agenda

Using human autonomy as an example, we argued that the discourse on bidirectional human-AI alignment would benefit from more theoretical clarity regarding the involved value concepts. We demonstrate how distinguishing material and experiential dimensions, multiple stakeholders, and lifecycle phases can extend and refine the bidirectional alignment framework. Supporting alignment requires identifying how structural conditions of the underlying values are established and distributed across stakeholders so that design decisions allow for the specific realization of these values. Without such operationalization, alignment faces the risk of remaining a normative call without translation into an implementable design practice, or only succeeding at a narrow definition of AI alignment.

At the same time, we caution that AI alignment for autonomy is not reducible to a set of measurable components, nor directly corresponds to a checklist implementable design strategies. How an AI system supports human autonomy needs to be reinterpreted for every context. Autonomy has been argued to function as a boundary object in HCI research; boundary objects are flexible yet robust concepts that enable collaboration across communities

of practice [5], being open to reinterpretation for a local context while at the same time providing a goal for orienting one's efforts. It is good to clarify and get a better handle on designing systems that respect human autonomy. However, supporting value-aligned action means something different in different contexts, where such contextual aspects may be missed when using a checklist approach.

This perspective provides multiple implications. Designers and researchers should explicitly specify which value dimensions they aim to support, and evaluate whether they are realized materially (through controllable outcomes) and experientially (e.g., through perceived liberty). Bidirectional alignment efforts should evaluate how feedback mechanisms or adaptations are experienced by different stakeholders. In this context, we propose to research related questions such as:

- How can abstract values be systematically decomposed and operationalized to be included in technical approaches for bidirectional alignment?
- Which methods allow for local reinterpretation of what supporting human autonomy means, when an AI system is applied to a new context or problem?
- How can we identify whether a value is sufficiently operationalized?
- How can alignment account for contradicting values of different stakeholders?
- In which lifecycle phases are alignment interventions most effective, and how do early decisions constrain later alignment efforts?

Through unpacking human autonomy and strategies to support it in the context of (bidirectional) alignment, we hope to contribute to a more grounded understanding and implementation of human-centered AI systems.

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